BLUE: 1-2-3-4 cycle; 1-2 is constant entropy. RED DASHED: pressure / enthalpy read-off guides.

R22 | Chiller

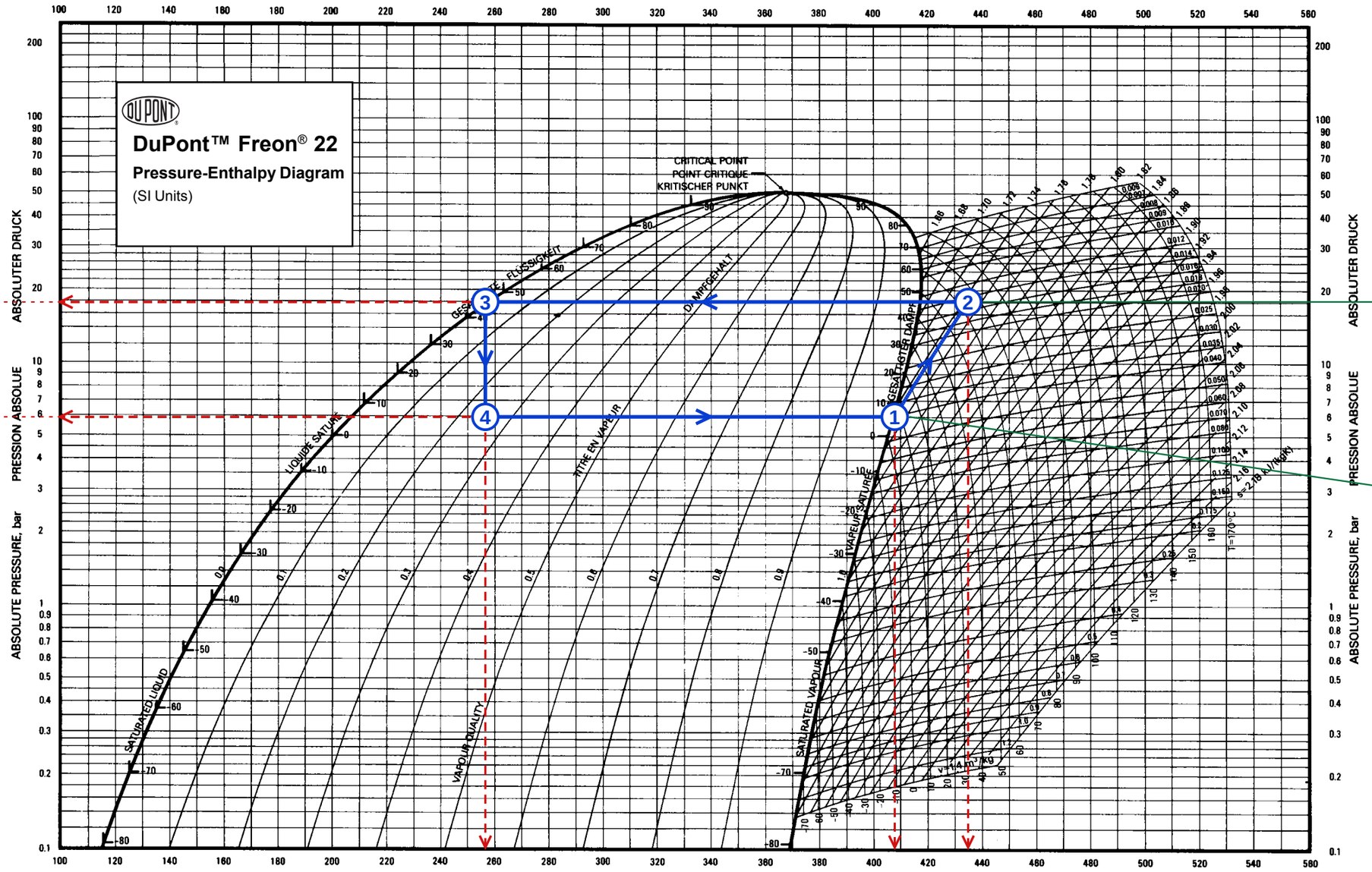
Te=5 degC; Tc=45 degC

1 -> 2: s1 = s2

Read v1 and v2 at points 1 and 2.

pc = 17.292
bar (absolute)

pe = 5.841
bar (absolute)



POINT 2: COMPRESSOR OUT
v2 = 0.01501 m3/kg
s2 = 1.74361 kJ/(kg.K)

POINT 1: COMPRESSOR IN
v1 = 0.04034 m3/kg
s1 = 1.74361 kJ/(kg.K)

h3=h4 = 256.36

h1 = 406.85

h2 = 433.82

FROM h READINGS
qe = h1-h3 = 150.49
w = h2-h1 = 26.97
COP = qe/w = 5.58
qe and w: kJ/kg

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

BLUE: 1-2-3-4 cycle; 1-2 is constant entropy. RED DASHED: pressure / enthalpy read-off guides.

R22 | Room cooler

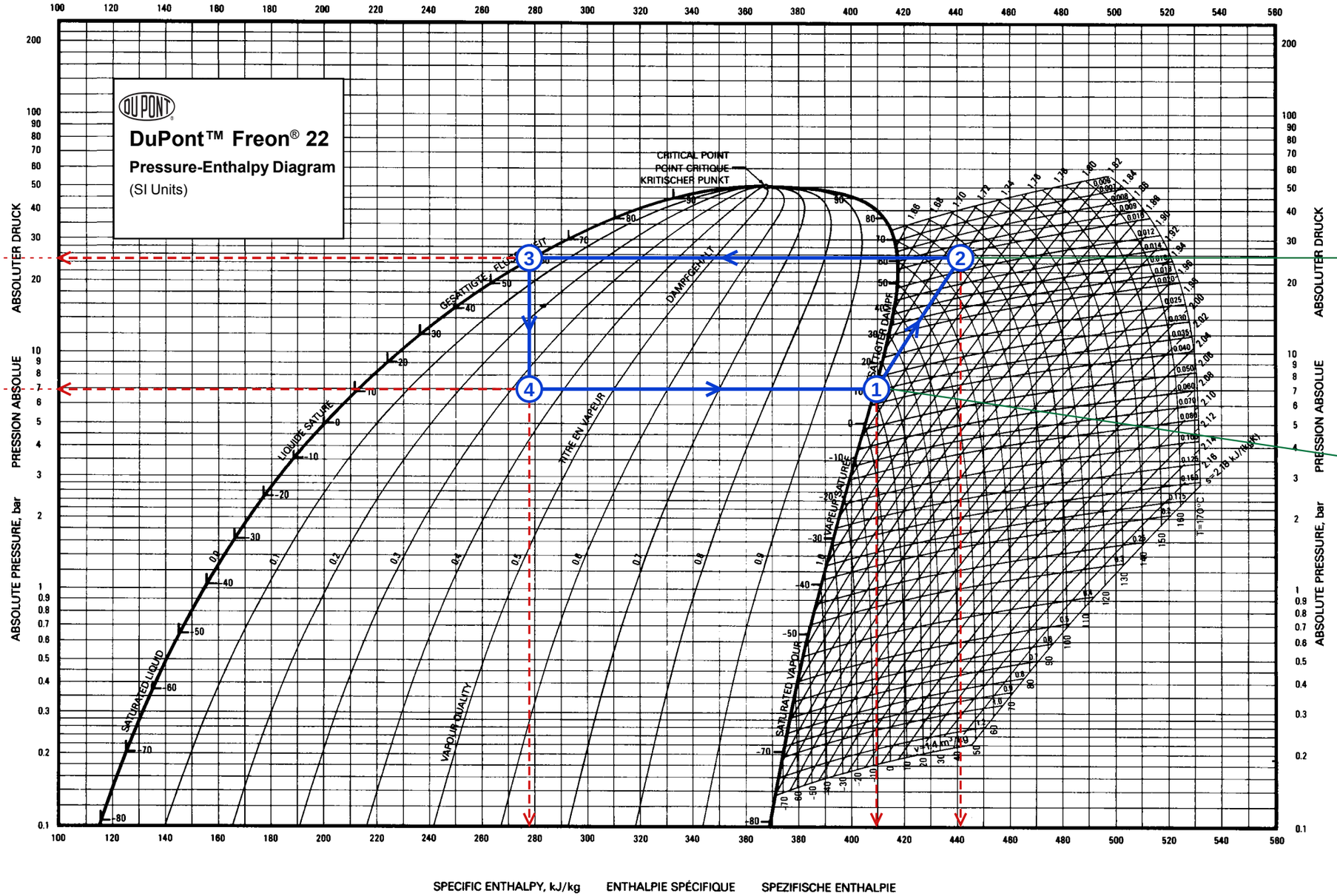
Te=10 degC; Tc=60 degC

1 -> 2: s1 = s2

Read v1 and v2 at points 1 and 2.

pc = 24.275
bar (absolute)

pe = 6.809
bar (absolute)



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h3=h4 = 277.61

h1 = 408.56

h2 = 440.29

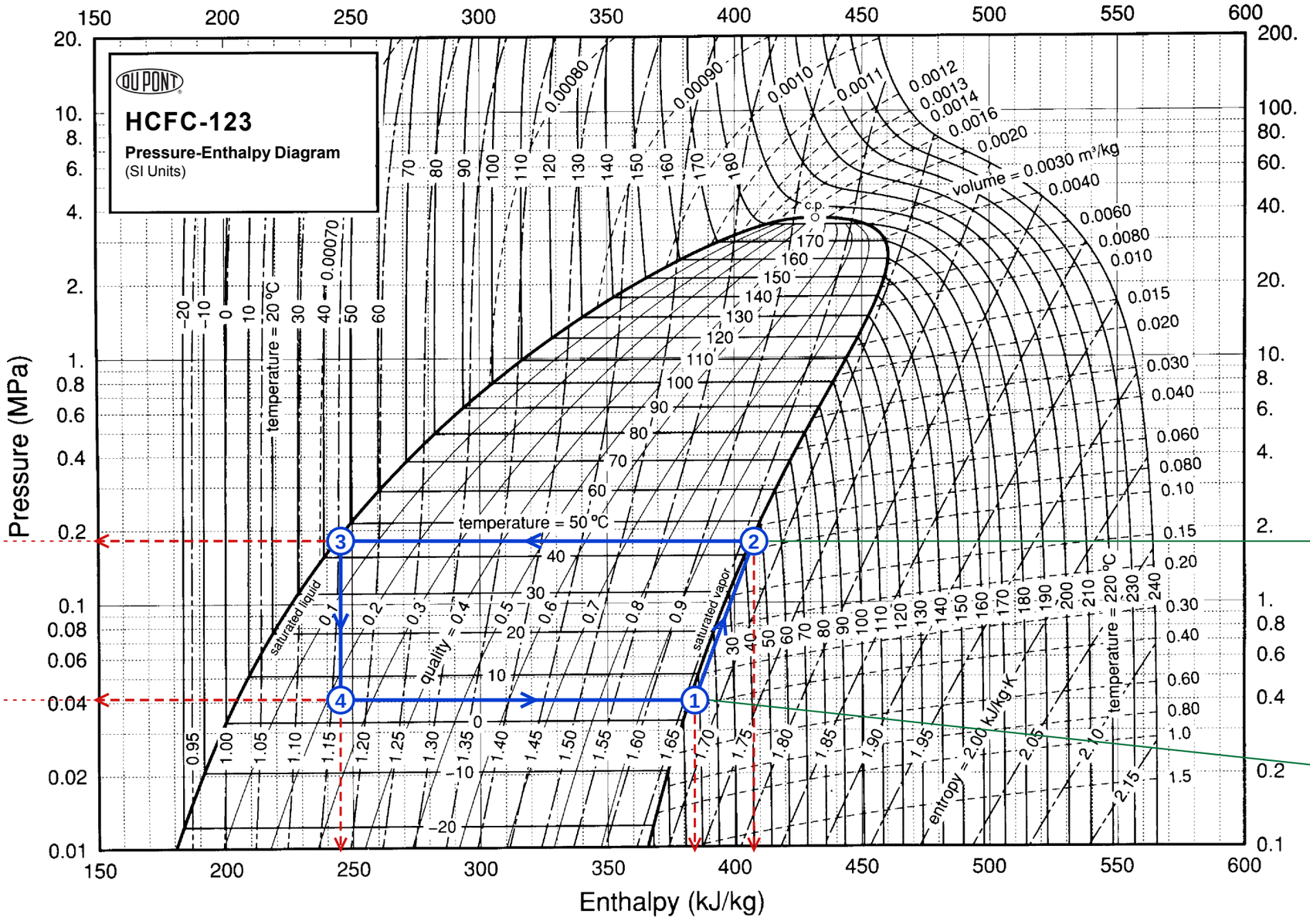
POINT 2: COMPRESSOR OUT
v2 = 0.01071 m3/kg
s2 = 1.73680 kJ/(kg.K)

POINT 1: COMPRESSOR IN
v1 = 0.03470 m3/kg
s1 = 1.73680 kJ/(kg.K)

FROM h READINGS
qe = h1-h3 = 130.95
w = h2-h1 = 31.73
COP = qe/w = 4.13
qe and w: kJ/kg

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

BLUE: 1-2-3-4 cycle; 1-2 is constant entropy. RED DASHED: pressure / enthalpy read-off guides.



pc = 0.1817
MPa / 1.817 bar

pe = 0.0408
MPa / 0.408 bar

h3=h4 = 245.81

h1 = 384.44

h2 = 407.57

R123 | Chiller

Te=5 degC; Tc=45 degC

1 -> 2: s1 = s2

Read v1 and v2 at points 1 and 2.
Ideal point 2 is slightly
inside the saturation dome.

POINT 2: COMPRESSOR OUT
v2 = 0.08853 m3/kg
s2 = 1.66325 kJ/(kg.K)

POINT 1: COMPRESSOR IN
v1 = 0.36203 m3/kg
s1 = 1.66325 kJ/(kg.K)

FROM h READINGS
qe = h1-h3 = 138.63
w = h2-h1 = 23.13
COP = qe/w = 5.99
qe and w: kJ/kg

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

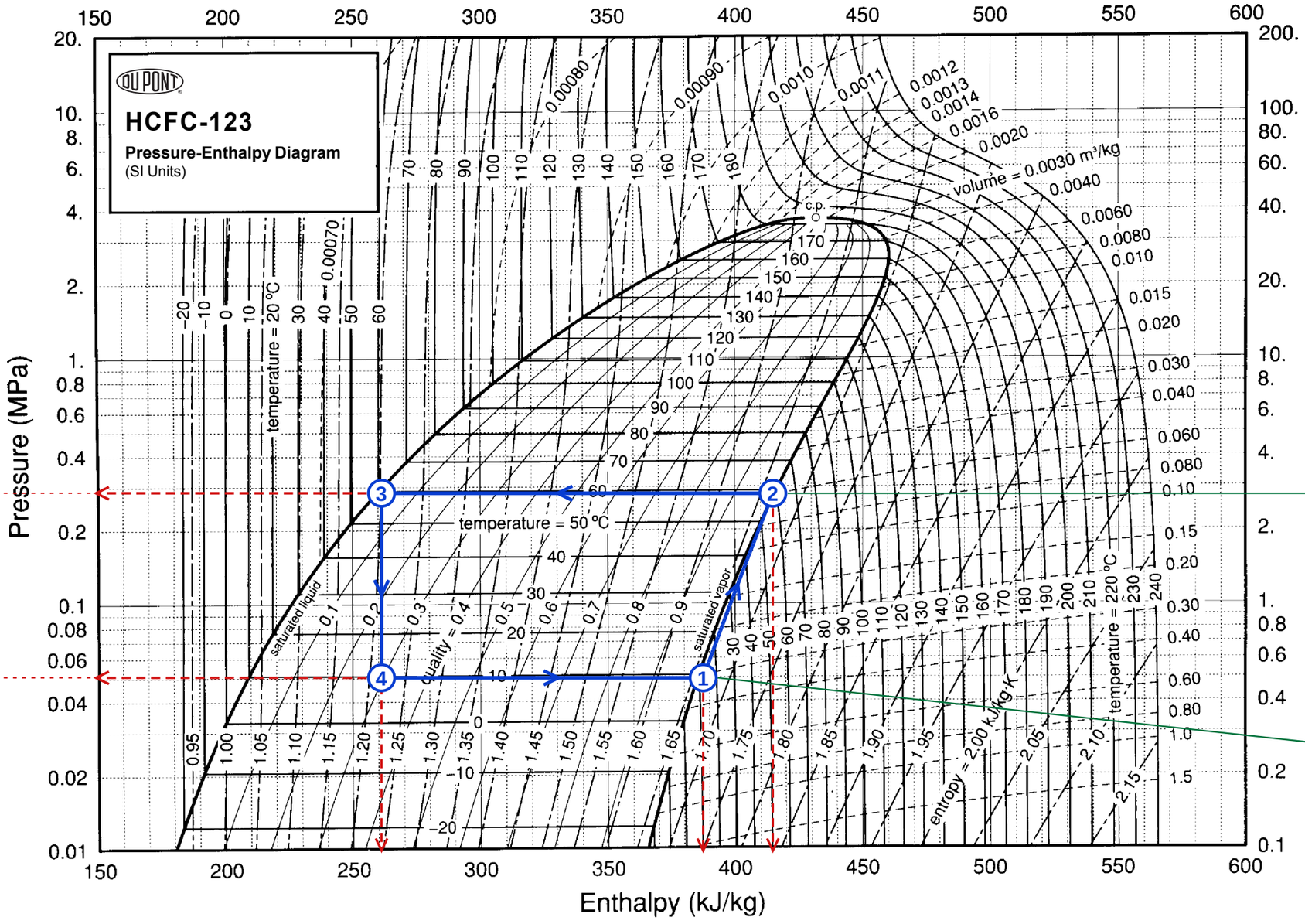
BLUE: 1-2-3-4 cycle; 1-2 is constant entropy. RED DASHED: pressure / enthalpy read-off guides.

R123 | Room cooler

Te=10 degC; Tc=60 degC

1 -> 2: s1 = s2

Read v1 and v2 at points 1 and 2.
Ideal point 2 is slightly
inside the saturation dome.



pc = 0.2859
MPa / 2.859 bar

pe = 0.0506
MPa / 0.506 bar

POINT 2: COMPRESSOR OUT
v2 = 0.05678 m3/kg
s2 = 1.66264 kJ/(kg.K)

POINT 1: COMPRESSOR IN
v1 = 0.29637 m3/kg
s1 = 1.66264 kJ/(kg.K)

h3=h4 = 261.67

h1 = 387.46

h2 = 414.71

FROM h READINGS
qe = h1-h3 = 125.79
w = h2-h1 = 27.25
COP = qe/w = 4.62
qe and w: kJ/kg

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

BLUE: 1-2-3-4 cycle; 1-2 is constant entropy. RED DASHED: pressure / enthalpy read-off guides.

R134a | Chiller

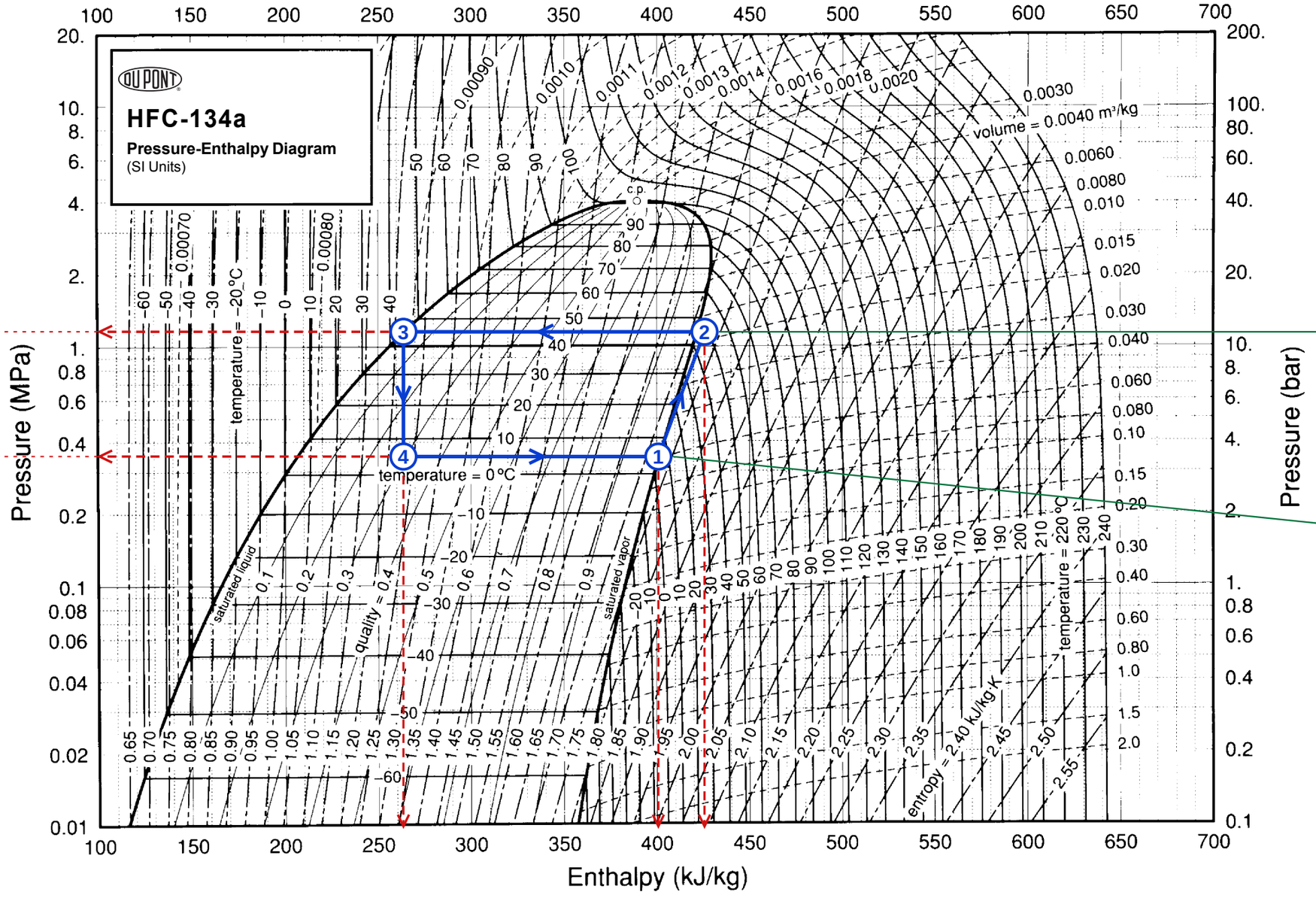
Te=5 degC; Tc=45 degC

1 -> 2: s1 = s2

Read v1 and v2 at points 1 and 2.

pc = 1.1599
MPa / 11.599 bar

pe = 0.3497
MPa / 3.497 bar



h3=h4 = 263.94

h1 = 401.49

h2 = 426.40

POINT 2: COMPRESSOR OUT
v2 = 0.01790 m3/kg
s2 = 1.72446 kJ/(kg.K)

POINT 1: COMPRESSOR IN
v1 = 0.05837 m3/kg
s1 = 1.72446 kJ/(kg.K)

FROM h READINGS
qe = h1-h3 = 137.55
w = h2-h1 = 24.91
COP = qe/w = 5.52
qe and w: kJ/kg

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

BLUE: 1-2-3-4 cycle; 1-2 is constant entropy. RED DASHED: pressure / enthalpy read-off guides.

R134a | Room cooler

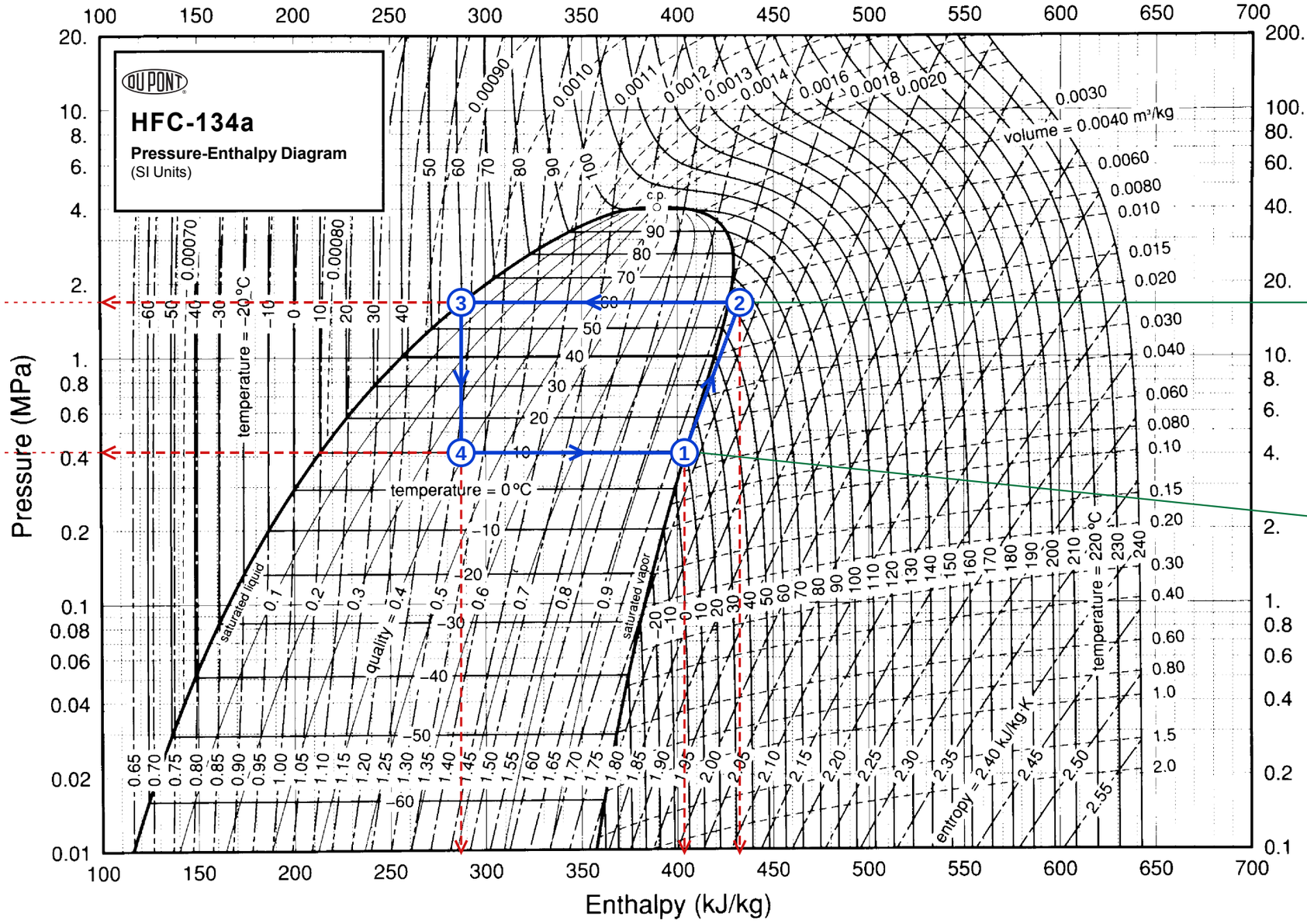
Te=10 degC; Tc=60 degC

1 -> 2: s1 = s2

Read v1 and v2 at points 1 and 2.

pc = 1.6818
MPa / 16.818 bar

pe = 0.4146
MPa / 4.146 bar



POINT 2: COMPRESSOR OUT
v2 = 0.01200 m3/kg
s2 = 1.72211 kJ/(kg.K)

POINT 1: COMPRESSOR IN
v1 = 0.04944 m3/kg
s1 = 1.72211 kJ/(kg.K)

h3=h4 = 287.50

h1 = 404.32

h2 = 433.24

FROM h READINGS
qe = h1-h3 = 116.82
w = h2-h1 = 28.92
COP = qe/w = 4.04
qe and w: kJ/kg

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

BLUE: 1-2-3-4 cycle; 1-2 is constant entropy. RED DASHED: pressure / enthalpy read-off guides.

R407C | Chiller

Te=5 degC; Tc=45 degC

1 -> 2: s1 = s2

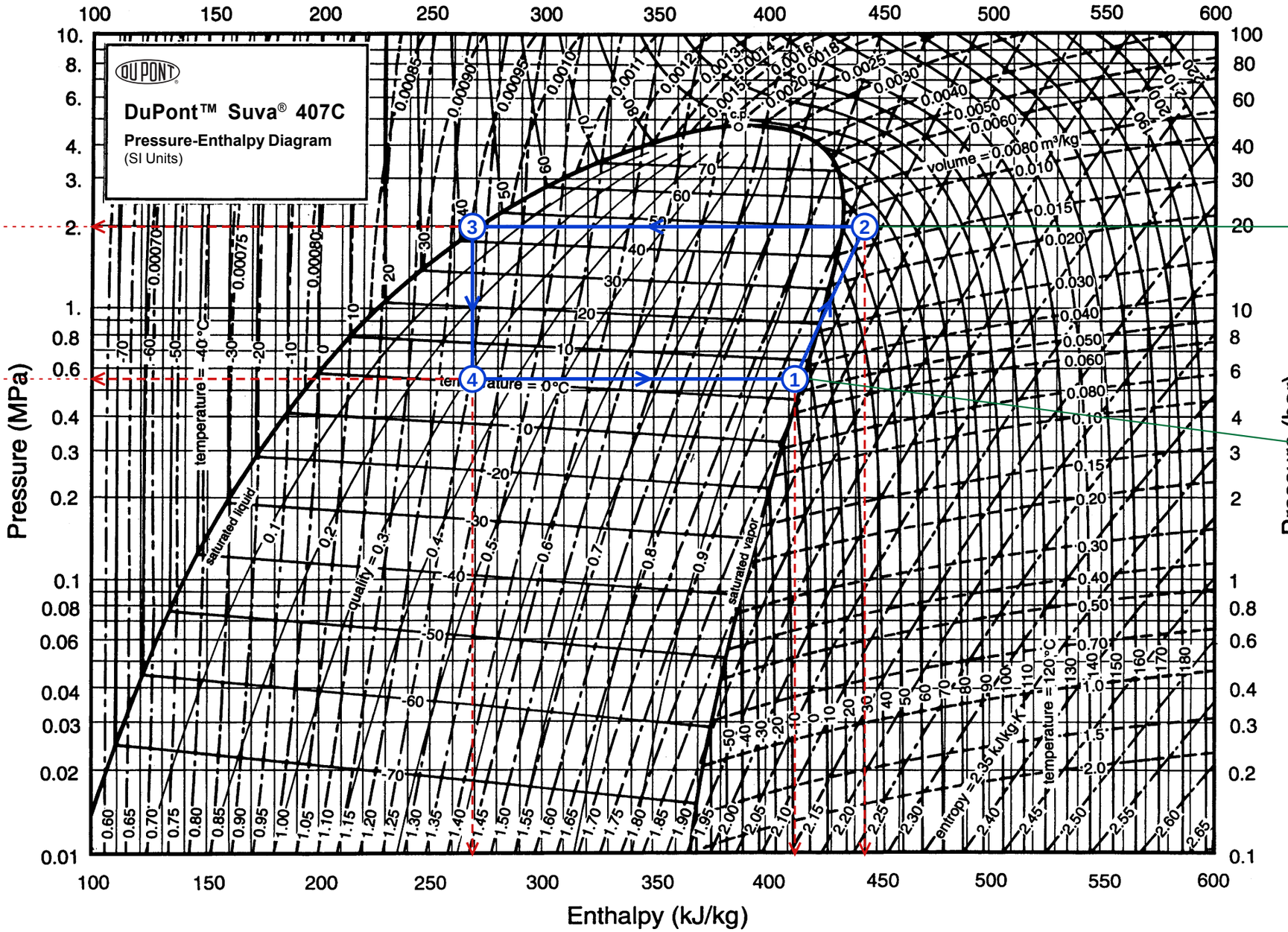
Read v1 and v2 at points 1 and 2.

Te uses dew point.

Tc uses bubble point.

pc = 1.9722
MPa / 19.722 bar

pe = 0.5469
MPa / 5.469 bar



POINT 2: COMPRESSOR OUT
v2 = 0.01241 m3/kg
s2 = 1.77001 kJ/(kg.K)

POINT 1: COMPRESSOR IN
v1 = 0.04291 m3/kg
s1 = 1.77001 kJ/(kg.K)

FROM h READINGS
qe = h1-h3 = 143.11
w = h2-h1 = 31.02
COP = qe/w = 4.61
qe and w: kJ/kg

h3=h4 = 268.64

h1 = 411.75

h2 = 442.77

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.

BLUE: 1-2-3-4 cycle; 1-2 is constant entropy. RED DASHED: pressure / enthalpy read-off guides.

R407C | Room cooler

Te=10 degC; Tc=60 degC

1 -> 2: s1 = s2

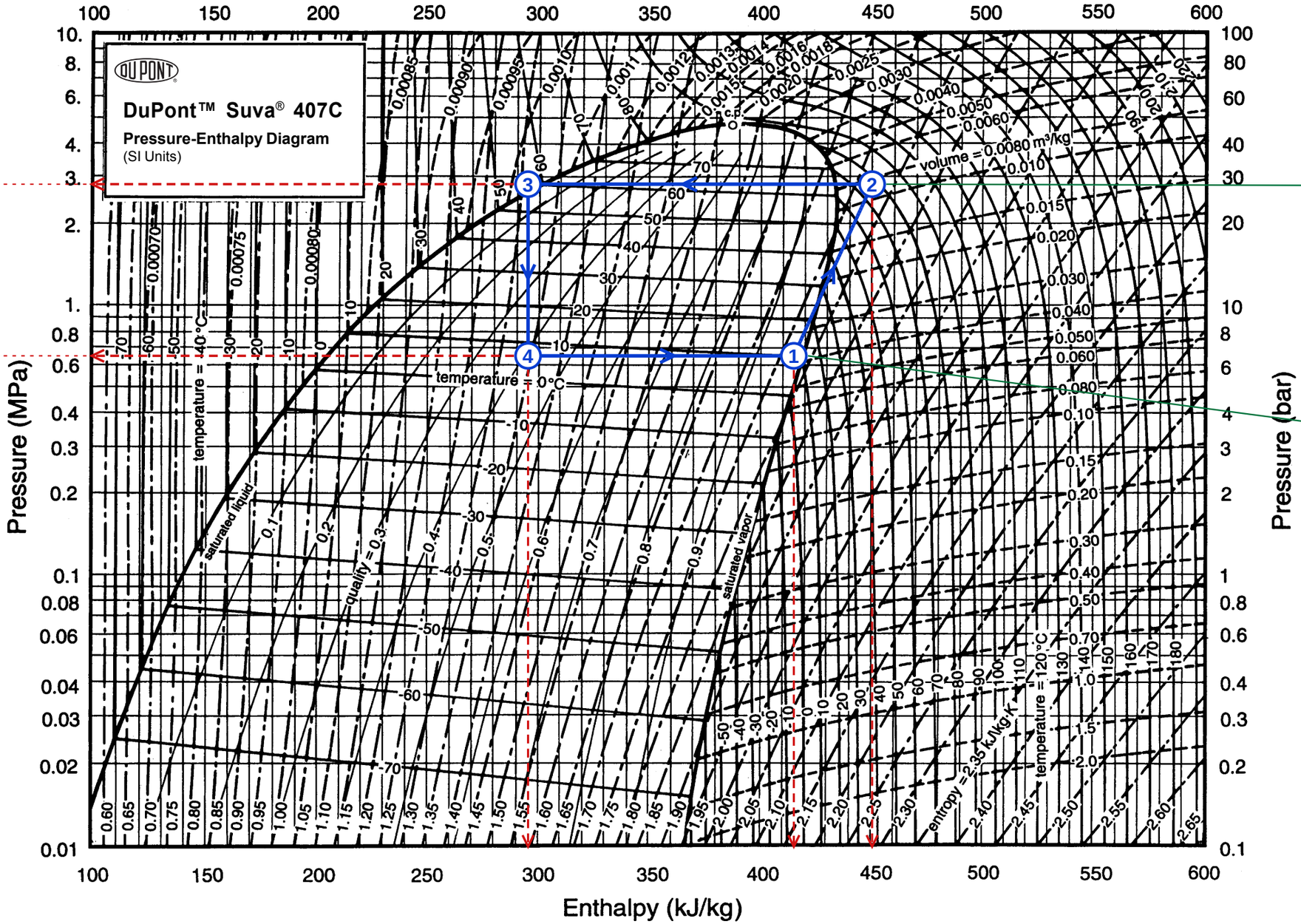
Read v1 and v2 at points 1 and 2.

Te uses dew point.

Tc uses bubble point.

pc = 2.7694
MPa / 27.694 bar

pe = 0.6449
MPa / 6.449 bar



POINT 2: COMPRESSOR OUT
v2 = 0.00864 m3/kg
s2 = 1.76430 kJ/(kg.K)

POINT 1: COMPRESSOR IN
v1 = 0.03643 m3/kg
s1 = 1.76430 kJ/(kg.K)

FROM h READINGS
qe = h1-h3 = 118.88
w = h2-h1 = 35.02
COP = qe/w = 3.39
qe and w: kJ/kg

h3=h4 = 295.14

h1 = 414.02

h2 = 449.04

h in kJ/kg. All p are absolute. Points are property-checked approximations; source-chart scan has finite reading precision.